The Analytics, Assets, and Performance Branch has received a data request from the Deputy Chief of the Maintenance Operations Administration. The deputy chief wants to know the following:

- Blocks with a count of excavation or paving permit renewals that expire after October 1st, 2025.
- Blocks with count of open roadway operations service requests created after October 1st, 2025.
- The latest expiration date from all the permits on the block.
- The latest creation date from all the open service request on the block.
- Blocks with these permits and open service requests.

He wants to see the permits, the service requests, and the roadway subblock centerline on a map so he can explore the data. One issue he's noticed in the past: some permit data is not snapped to the roadway. He needs you to generate a version of the permit data that is snapped correctly.

The Deputy Chief of the Maintenance Operations Administration has asked for the results in an ArcGIS Pro project file or a QGIS project file.

You will enrich the Roadway Subblock feature with permit data, service request data, and dates. You will also create a version of the permit data snapped to the roadway.

Data Requirements

Filter construction permits using the following criteria:

- Permit expiration dates should be equal to or after October 1st, 2025.
- Only *renewal* permits that have been *issued*.
- Only excavation and paving permits.

Filter the roadway subblock data using the following criteria:

- Only streets, service lanes, and ramps.

Filter the Cityworks service requests using the following criteria:

- The request creation dates should be equal to or after October 1st, 2025.
- Requests should be open, pending, or have a completed inspection.
- Only roadway repair, pothole, or paving requests.

There is no data dictionary available. You will need to explore the data and determine how best to handle this project. We have provided the SQL definition query for the Cityworks Service Request

Data Sources

Construction Permit data is available via a REST Services feed here:
https://maps2.dcgis.dc.gov/dcgis/rest/services/DCGIS_APPS/DDOT_TOPS/MapServer/23

Roadway Subblock is available via a REST Services feed here:
https://maps2.dcgis.dc.gov/dcgis/rest/services/DCGIS_DATA/Transportation_WebMercator/MapServer/162

Cityworks Service Requests
https://maps2.dcgis.dc.gov/dcgis/rest/services/DDOT/Cityworks/MapServer/1

> STATUS NOT IN ('CLOSED', 'CANCEL', 'COMPLETE') And DESCRIPTION IN ('ROADWAY REPAIR', 'POTHOLES', 'STREET REPAVING', 'STREET REPAIR', 'STREET PAVING SCHEDU') And INITIATEDDATE >= timestamp '2025-10-01 00:00:00'

Deliverables Summary

- The **packaged ArcGIS Pro project file** or the **packaged QGIS project file** or a **.zip file of the project directory**. The project should contain the following as a *minimum* requirement:
    - A local geodatabase version of the Roadway Subblock feature class that has been enriched with
        - Count of permits on each subblock
        - Max expiration date among all permits on subblock
        - Count of service requests on each subblock
        - Max initiate date among all service requests on subblock
        - Flags for the blocks with both permits and service requests
    - A local geodatabase version of the filtered construction permit data. This data should be snapped to the roadway
    - The REST services feeds for permits, cityworks requests, and roadway subblock loaded as layers on the map.
    - The geoprocessing history.
- Include additional features classes, tables, files, notebooks, or python scripts you used to complete the assignment.

- This deliverable does not need finalized symbology. **Do not worry about spending time on a final polished design. We prefer to see your geoprocessing work.**
- All files should be compressed into a parent .zip folder with the following name,
    - *FirstName_MiddleInitial_LastName*_GIS.zip

Additional Guidance

If you get stuck and run out of time, don't worry! Partially completed projects are fine to submit. You are also welcome to submit narrative documents explaining your approach and how you planned to handle any issues with more time. We would rather see incomplete work than nothing at all.

Use of AI is expected. District government only has access to Copilot, but you may use any AI tool to complete your work. All these data sources are publicly available.